Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: Agentic AI for Scheduling | *Python, PyTorch, LLM Agents, RAG, CP-SAT, FastAPI*

- Built an agentic AI assistant that resolves free-text requests end to end: it reads live state through typed tools, drafts changes, verifies them with a CP-SAT solver and negotiates with affected stakeholders before anything is committed.
- Fine-tuned Qwen3.5-4B (LoRA, SFT + DPO in PyTorch) on agent traces distilled from Gemini; the 4B model **outscored its Gemini teacher** (17/18 vs 15/18) and passed **190/190** scenarios across five test suites.
- Compared agentic and fixed-pipeline orchestration on cost, determinism and error recovery (∼16K vs ∼200 tokens per request), and served both via FastAPI with streamed traces for observability.
- Built hybrid BM25 + dense RAG over OCR-ingested policy documents (**93.4%** Recall@5), with cited-rule denials.
- Designed responsible-AI guardrails (authority checks, cited rules, human approval): **0 of 50** adversarial attacks succeeded.

Retrieve Ten, Keep Three: Re-ranking RAG Evaluation | *Python, FAISS, Cross-Encoders, BM25, FastAPI*

- Built a two-stage RAG over the GDPR (bi-encoder + FAISS retrieves 10, cross-encoder keeps 3), raising Hit@3 from 0.46 to **0.85** on hand-labelled stress questions for **83 ms** per query (under 1% of end-to-end latency).
- Fused BM25 and dense retrieval with RRF, lifting Hit@10 from 0.77 to **0.93** on 100 generated questions for 1.3 ms; with a stronger re-ranker, LLM-judged correct answers rose from 61% to **75%**.
- Built a Qwen3-14B answer judge (**0.87** agreement with hand labels) and audited the test set, showing most of one corpus's wrong answers came from mislabelled gold.
- Deployed a FastAPI + React app on Railway with ONNX Runtime encoders to upload PDFs and compare dense, BM25 and fused ranks per chunk.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, SentencePiece, ONNX Runtime Web*

- Built Hindi and Assamese LMs end to end: curated 1.25B tokens (20%+ self-crawled and OCR'd), trained native 16K BPE tokenizers, and pretrained four 25M-param Transformers from PyTorch primitives on 500M tokens each.
- Fine-tuned on solver-generated reasoning, reaching **0.96** test accuracy and **0.85** on 10-hop chains (3× the training depth) vs a 0.48 baseline, with a leakage check ruling out answer lookup.
- Exported all 10 models to ONNX with a KV cache and a JavaScript tokenizer so they run fully in the browser, deployed as a live demo on Railway via Docker.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Built a BitTorrent-style P2P system in C++; self-scheduling workers cut 1GB single-peer downloads from **193s to under 13s**.
- Designed a pluggable piece-selection interface (rarest-first, sequential, random) chosen at runtime, with out-of-order parallel chunk writes to exact file offsets.
- Architected Primary-Backup tracker failover with race-free election and state sync; **46 unit tests** pass under ASan/UBSan.

Technical Skills

Languages: Python, C/C++, TypeScript, JavaScript, SQL, Bash.

Agentic AI & LLMs: Multi-Agent Systems, Tool Calling, RAG, Re-ranking, Prompt Engineering, Guardrails, LoRA.

Machine Learning: PyTorch, Transformers, Hugging Face, SentencePiece, ONNX, FAISS, Scikit-learn, NumPy.

Software Engineering: Data Structures & Algorithms, OOP, FastAPI, REST APIs, pytest, Git, Docker, CI/CD.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast; Chess and Badminton.